EE3032 Introduction to VLSI Design—Homework 4

1. A CMOS full adder with the inputs, A_i, B_i, and C_i, and outputs, C_{i+1} and S_i. One can express the Boolean function of the full adder as follows:

$$\begin{aligned} C_{i+1} &= A_iB_i + B_iC_i + A_iC_i \\ S_i &= (A_i + B_i + C_i)\overline{C}_{i+1} + A_iB_iC_i \end{aligned}$$

Also, the full adder has the symmetric feature. Therefore, one can design a CMOS full adder using the transitor diagram shown in Fig. 1.

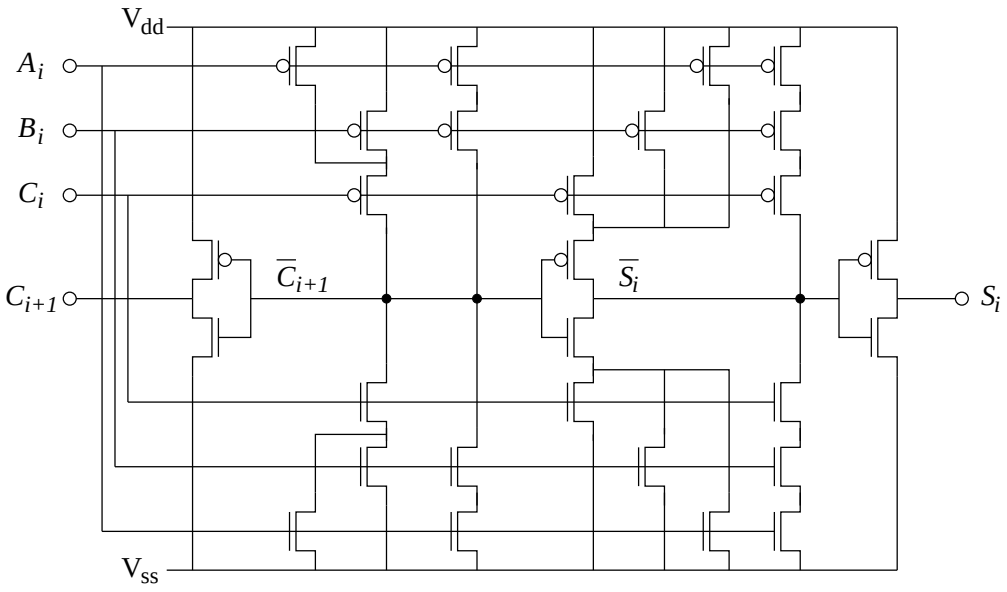

Figure 1: Mirror CMOS full adder.

Complete the following questions:

(a) Design the circuit using TSMC $0.18\mu m$ technology such that the $t_r = t_f$. Verify the circuit and measure its worst-case propagation delay using Hspice. (5 points)

(b) Realize the full adder using Domino logic using TSMC $0.18\mu m$ technology such that the $t_r = t_f$. Verify your design and measure its worst-case propagation delay using Hspice. (10 points)